BCBS 239: Principles for effective risk data aggregation and risk reporting

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Introduction

(1)) One of the most significant lessons learned from the global financial crisis that began in 2007 was that banks' information technology (IT) and data architectures were inadequate to support the broad management of financial risks. Many banks lacked the ability to aggregate risk exposures and identify concentrations quickly and accurately at the bank group level, across business lines and between legal entities. Some banks were unable to manage their risks properly because of weak risk data aggregation capabilities and risk reporting practices. This had severe consequences to the banks themselves and to the stability of the financial system as a whole.

(2) In response, the Basel Committee issued supplemental Pillar 2 (supervisory review process) guidance to enhance banks' ability to identify and manage bank-wide risks. In particular, the Committee emphasised that a sound risk management system should have appropriate management information systems (MIS) at the business and bank-wide level. The Basel Committee also included references to data aggregation as part of its guidance on corporate governance.

(3) Improving banks' ability to aggregate risk data will improve their resolvability. For global systemically important banks (G-SIBs) in particular, it is essential that resolution authorities have access to aggregate risk data that complies with the FSB's Key Attributes of Effective Resolution Regimes for Financial Institutions as well as the principles set out below. For recovery, a robust data framework will help banks and supervisors anticipate problems ahead. It will also improve the prospects of finding alternative options to restore financial strength and viability when the firm comes under severe stress. For example, it could improve the prospects of finding a suitable merger partner.

Definition

(8) For the purpose of this paper, the term “risk data aggregation” means **defining, gathering and processing risk data** according to the bank's risk reporting requirements to enable the bank to measure its performance against its risk tolerance/appetite. This includes sorting, merging or breaking down sets of data.

Objectives

(9) This paper presents a set of principles to strengthen banks' risk data aggregation capabilities and internal risk reporting practices (the Principles). In turn, effective implementation of the Principles is expected to enhance risk management and decisionmaking processes at banks.

Overarching governance and infrastructure

(26) A bank should have in place a strong governance framework, risk data architecture and IT infrastructure. These are preconditions to ensure compliance with the other Principles included in this document. In particular, a bank's board should oversee senior management's ownership of implementing all the risk data

aggregation and risk reporting principles and the strategy to meet them within a timeframe agreed with their supervisors.

Principle 1 Governance

A bank's risk data aggregation capabilities and risk reporting practices should be subject to strong governance arrangements consistent with other principles and guidance established by the Basel Committee.

(27) A bank's board and senior management should promote the identification, assessment and management of data quality risks as part of its **overall risk management framework**.

(28) A bank's board and senior management should **review and approve** the bank's group risk data aggregation and risk reporting framework.

(29) A bank's risk data aggregation capabilities and risk reporting practices should be:

- (a) **Fully documented** and subject to **high standards of validatio**n. This validation should be independent and review the bank's compliance with the Principles in this document. Common practices suggest that the independent validation of risk data aggregation and risk reporting practices should be conducted using staff with specific IT, data and reporting expertise.
- (b) When **considering a material acquisition**, a bank's due diligence process should assess the risk data aggregation capabilities and risk reporting practices of the acquired entity, as well as the impact on its own risk data aggregation capabilities and risk reporting practices.
- (c) In particular, risk data aggregation capabilities should be independent from the choices a bank makes regarding its legal organisation and geographical presence.

(30) A bank's senior management should be fully aware of and understand the limitations that prevent full risk data aggregation, in terms of coverage (eg risks not captured or subsidiaries not included), in technical terms (eg model performance indicators or degree of reliance on manual processes) or in legal terms (legal impediments to data sharing across jurisdictions).

Principle 2 Data architecture and IT infrastructure

A bank should design, build and maintain data architecture and IT infrastructure which fully supports its risk data aggregation capabilities and risk reporting practices not only in normal times but also during times of stress or crisis, while still meeting the other Principles.

Risk data aggregation capabilities

Banks should develop and maintain strong risk data aggregation capabilities to ensure that risk management reports reflect the risks in a reliable way (ie meeting data aggregation expectations is necessary to meet reporting expectations). Compliance with these Principles should not be at the expense of each other. These risk data aggregation capabilities should meet all Principles below simultaneously in accordance with paragraph 22 of this document.

Principle 3 Accuracy and Integrity

A bank should be able to generate accurate and reliable risk data to meet normal and stress/crisis reporting accuracy requirements. Data should be aggregated on a largely automated basis so as to minimise the probability of errors.

(36) A bank should aggregate risk data in a way that is accurate and reliable.

- (c) Risk data should be reconciled with bank's sources, including accounting data where appropriate, to ensure that the risk data is accurate.
- (d) A bank should strive towards a single authoritative source for risk data per each type of risk.

(37) As a precondition, a bank should have a "dictionary" of the concepts used, such that data is defined consistently across an organisation.

(38) There should be an appropriate balance between automated and manual systems. Where professional judgements are required, human intervention may be appropriate.

Principle 4 Completeness

A bank should be able to capture and aggregate all material risk data across the banking group. Data should be available by business line, legal entity, asset type, industry, region and other groupings, as relevant for the risk in question, that permit identifying and reporting risk exposures, concentrations and emerging risks.

(43) Where risk data is not entirely complete, the impact should not be critical to the bank's ability to manage its risks effectively. Supervisors expect banks' data to be materially complete, with any exceptions identified and explained.

Principle 5 Timeliness

A bank should be able to generate aggregate and up-to-date risk data in a timely manner while also meeting the principles relating to accuracy and integrity, completeness and adaptability. The precise timing will depend upon the nature and potential volatility of the risk being measured as well as its criticality to the overall risk profile of the bank. The precise timing will also depend on the bank-specific frequency requirements for risk management reporting, under both normal and stress/crisis situations, set based on the characteristics and overall risk profile of the bank.

(44) A bank's risk data aggregation capabilities should ensure that it is able to produce aggregate risk information on a timely basis to meet all risk management reporting requirements.

Principle 6 Adaptability

A bank should be able to generate aggregate risk data to meet a broad range of on-demand, ad hoc risk management reporting requests, including requests during stress/crisis situations, requests due to changing internal needs and requests to meet supervisory queries.

(48) A bank's risk data aggregation capabilities should be flexible and adaptable to meet ad hoc data requests, as needed, and to assess emerging risks. Adaptability will enable banks to conduct better risk management, including forecasting information, as well as to support stress testing and scenario analyses.